

Sprint Action Plan: Person 2: Compute Lead

T4 Latency Profiling, Peak VRAM & Scaling Exponents • Adaptive Attention Computation Sprint

Assigned Role: Person 2: Compute Lead	Estimated Workload: 9.5 Hours	Sprint Timeline: 0.0h – 9.5h
Core Discipline: CUDA Performance Benchmarking / Systems Profiling	Primary Handoff To: Person 6 (Synthesis Lead)	Target Hardware: Dual T4 (16GB)

1. Role Mission & Strategic Objective

Profile execution time, peak memory, and empirical scaling exponents on Kaggle Tesla T4 across sequence lengths $N = 64$ to $65,536$. Establish the crossover point between Softmax, SDPA, and Linear Attention while enforcing strict OOM guardrails.

2. Input Dependencies & Final Deliverables

Type	Item / Specification	Source / Destination	Deadline / Gate
INPUT	Frozen CSV schema contract	Person 6 (Synthesis Lead)	Hour 1.0
INPUT	Linear and mock attention kernels	Person 1 (Theory Lead)	Hour 1.0
OUTPUT	T4 Efficiency Notebook (01_T4_Efficiency.ipynb)	Team Repository	Hour 7.5
OUTPUT	Raw latency and memory logs (efficiency_raw.csv)	Person 6 (Synthesis Lead)	Hour 7.5
OUTPUT	High-res efficiency scaling charts (Fig 1)	Person 6 (Synthesis Lead)	Hour 8.0

3. Technical Specifications & Contract Signature

Key Code / Mathematical Contract for Person 2: Compute Lead:

- SDPA Backend Flags: `torch.backends.cuda.enable_mem_efficient_sdp(True); enable_flash_sdp(False)`
- Timing Harness: Use CUDA Events (`torch.cuda.Event(enable_timing=True)`) with explicit synchronization
- Memory Tracking: Measure $(\text{max_memory_allocated}() - \text{base_memory}) / (1024^2)$ in MB
- Sweep Regimes: $N \in \{64, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768, 65536\}$; $B=1, H=4, d=64$

4. Hour-by-Hour Master Action Checklist — Person 2: Compute Lead

Window	Specific Actions & Step-by-Step Instructions	Deliverable / Evidence	Status
0.0 – 1.0 h	Setup Kaggle dual T4 harness; configure CUDA event timing harness; lock CSV output schema.	Benchmarking harness template	[] Done
1.0 – 3.0 h	Benchmark Naive Softmax vs SDPA across N = 64 to 32,768; record exact OOM boundary at N = 32,768.	Softmax & SDPA latency/memory curves	[] Done
3.0 – 5.5 h	Benchmark ReLU+1 Linear Attention latency and peak VRAM up to N = 65,536; confirm linear scaling slope.	Linear attention scaling curves	[] Done
5.5 – 7.5 h	Fit empirical scaling exponents (log-log slope α); calculate median speedups relative to Softmax and SDPA.	efficiency_raw.csv + scaling exponents	[] Done
7.5 – 8.5 h	Export publication-quality 3-panel scaling figures (latency, memory, speedup) to Person 6.	Rendered PNG charts delivered to P6	[] Done
8.5 – 9.5 h	Verify end-to-end Kaggle reproduction script executes without interactive timeouts.	Verified Kaggle reproduction notebook	[] Done

5. Operational Guardrails & Failure Protocols

Constraint / Risk	Enforcement Rule & Exact Action
CUDA OOM Crashes	Enforce strict batch size B=1 for all sequence lengths $N \geq 8,192$.
Asynchronous Timing Distortion	Always call torch.cuda.synchronize() before start and after end events.
Kaggle 60-min Idle Timeout	Submit long sweeps ($N \geq 16,384$) via 'Save & Run All (Commit)'.

6. Definition of Done (DoD) & Sign-Off Checklist

[]	Latency measured up to N = 65,536 (confirming 9.74 ms for linear attention).
[]	Naive softmax OOM point documented at N = 32,768.
[]	efficiency_raw.csv populated with median, Q25, Q75, and peak VRAM.
[]	Scaling exponents α reported (0.80 for linear vs 1.81 for softmax).